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Antibiotyping and Genotyping of Extensively Drug Resistant (XDR) *Salmonella* Spp. Isolated from Clinical Samples of Lahore, Pakistan.

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Running headline: Antibiotyping and Genotyping of XDR *Salmonella*.

Abstract:

Aims: Antibiotic resistance is a major problem in *Salmonella enterica* serovar Typhi. Objective of this study was to evaluate the prevalence of XDR *Salmonella* among local population of Lahore and genotyping of isolates for antibiotic resistant genes. **Methods and Results:** A total of 200 blood samples from suspected typhoid fever patients were collected. One hundred and fifty-seven bacterial samples were confirmed as *Salmonella* Typhi and twenty-three samples were confirmed as *Salmonella* Paratyphi after biochemical, serological and PCR based molecular characterization.

Antibiogram analysis classified 121 (67.2%) *Salmonella* isolates as MDR and 62 isolates (34.4%) as XDR. The predominant resistance gene was *ampC* with 47.7% prevalence, followed by *gyrA, catA1, tet(A), aac (3)-la, qnrS, bla*$_{NDM-1}$ and *bla*$_{CTX-M-15}$ genes in 45.5%, 40%, 21.6%, 18.3%, 11.6%, 2.2% and 0.5% isolates, respectively. Sequence analysis showed the presence of *sul1* and *dfrA7* gene cassette arrays in twelve class 1 integron integrase positive isolates. **Conclusion:** Large number of clinical XDR *S.* Typhi resistant against third generation cephalosporins have been reported. **Significance and Impact of Study:** Current study highlights the possible emergence of clinical XDR *S.* Typhi cases in Lahore, Pakistan. Potential attribution of phenotypic and genotypic XDR cases may help to contribute targeted therapy.

Keywords: Extensively Drug Resistant; Multiple Drug Resistance; *Salmonella;* Enteric Fever; Pakistan.

Introduction:

Typhoid fever is caused by *Salmonella enterica* subsp. *enterica* serovar Typhi (*S.* Typhi) is a rod-shaped, Gram-negative facultative anaerobe that belongs to the family Enterobacteriaceae. *S.* Typhi transfer from human to human via fecal-oral route and by contaminated water. Typhoid fever is a severe public health concern specially in under-developed countries with 128,000-161,000 deaths each year (Crump *et al.* 2004; Chirico *et al.* 2020). During 2016-2017, more than 300 cases of extensively drug resistant (XDR) *S.* Typhi strain were reported in Pakistan (Klemm *et al.* 2018). In response to the XDR cases from Pakistan, 29 cases of typhoid fever were reported in USA which had travel history from Pakistan (Chatham-Stephens *et al.* 2019).

Improved sanitation, vaccination and responsible use of antibiotics are important factors to treat typhoid infection. But unfortunately, irrational use of antibiotics leads to develop resistance against first-line and second-line antibiotics and emergence of multiple drug resistant (MDR) bacterial spp. The first line antibiotic to treat typhoid infection are ampicillin, chloramphenicol and trimethoprim-sulfamethoxazole (Crump *et al.* 2015). Any *Salmonella* strains resistant to first-line antibiotics are termed as Multiple Drug Resistant (MDR). To treat MDR *Salmonella* infections, second-line antibiotics including ceftriaxone, fluoroquinolones, third generation cephalosporin and macrolide are now in use (Crump *et al.* 2015). Resistance to one or more second-line antibiotics could generate extensively drug resistant (XDR) *Salmonella*. XDR *Salmonella* is a serious local and global health concern due to its minimal treatment options. First case of ceftriaxone resistant *Salmonella* was reported in Hyderabad, Sindh, Pakistan in 2016 (Yousafzai *et al.* 2019). In Pakistan, almost 5.01% rise in XDR *Salmonella* cases were observed since 2015 (Saeed *et al.* 2019).

Microbes usually transfer antimicrobial resistance (AMR) genes among microbial communities by horizontal acquisition of resistance genes, activated through insertion sequences, transposons and conjugative plasmids, recombination of external DNA into the chromosome, or through mutations in various chromosomal loci (Ugboko and De 2014). Mobile genetic elements like transposons and integrons can disperse antimicrobial resistance genes (Firoozeh *et al.* 2011). In *S.* Typhi, antibiotic resistance is mostly mediated by the plasmids. Incompatibility group containing (*Inc*) HI1 plasmids are key antibiotic resistance vectors in *S.* Typhi (Wain *et al.* 2001). Class 1 integrons facilitate the acquisition of more than one gene cassettes in a specific recombination site (*attI1*) so consisting clumps of antibiotic resistance genes (Hall and Collis 1995). Class 1 integrons comprises a 5′-conserved segment (5′-CS) which contains the integrase *intI1* gene, the recombination region *attI1*, and the pc promoter. This class of integrons also consist of a 3′-conserved segment (3′-CS), which contains the genes *sul1* and *qacEΔ1* that encrypt resistance to sulfonamides and quaternary ammonium compounds, respectively (de Toro *et al.* 2011).

Carbapenems are a class of antimicrobial β-lactam elements that have a wide spectrum effect on Gram-negative bacteria. Carbapenems have been thought the last-resort antimicrobial agents for Gram-negative bacterial infections (Wang *et al.* 2017). There are three main categories of ESBL encoding genes: *blaCTX-M*, *blaSHV* and *blaTEM*. The category *blaCTX-M* has also been designated into five subgroups *blaCTX-M-1, blaCTX-M-2, blaCTX-M-8, blaCTX-M-9,* and *blaCTX-M-25* (ur Rahman *et al.* 2018). On the other side, resistance to ciprofloxacin is primarily due to double mutations in *gyrA* and only a single mutation in *parC* which is present in *Salmonella* (Hooper 2001). Quinolones, such as nalidixic acid, are antibiotics targeting topoisomerases of bacterial type II and, more precisely, DNA gyrase and DNA topoisomerase IV. Both proteins are encrypted by the genes *gyrA*, *gyrB* and *parC*, *parE* and modulate DNA supercoiling. Quinolones inhibit these enzymes, leading to disrupted replication of chromosomes and rapid bacterial expiries (Cuypers *et al.* 2018).

The acquisition of extended-spectrum β-lactamase *blaCTX-M-15* genes and *qnrS* fluoroquinolones resistance gene in *S.* Typhi, transforms multidrug resistant (MDR) strains to extensively drug resistant (XDR) strains. The situation become worse when the same bacterium (along with other features including plasmids, ICEs or certain phages) also harbors integrase *intI1* gene, making its antimicrobial resistant genes transfer among other bacteria. Although azithromycin resistance has been reported in XDR *Salmonella* strains prevalent in India, there are still two main treatment options left including the use of meropenem and azithromycin (Manesh *et al.* 2017).

Material and Methods

Sample collection and processing:

Blood samples of suspected enteric fever patients were collected from Chughtai's Laboratory, Lahore, Pakistan. Blood samples from sixty-seven patients were collected in vacutainer and stored at 4°C for further analysis at the Department of Microbiology, Faculty of Life Sciences, University of Central Punjab Lahore, Pakistan. Clinical representation and relevant investigation were recorded from each patient. Permission was obtained from the departmental ethical committee and informed consent was also taken from all the patients.

Bacterial isolation and identification:

Blood samples from enteric fever suspected patients were further processed for culture on blood and MacConkey agar. Purified bacterial isolates were identified based on colony morphology (Moore 1984), biochemical test using API 20E (BioMérieux, France) and serological reactions with *Salmonella* O antiserum factor 9, antiserum Factor 2 antisera, and antiserum H and Vi (Pro-Lab Diagnostics, Canada).

Antimicrobial susceptibility testing:

Antibiotic susceptibility testing was performed using a Kirby-Bauer method according to Clinical Laboratory Standard Institute (CLSI) guidelines (Wayne 2011) on Mueller-Hinton agar (Difco, USA). The following 09 antibiotic disks were used: ampicillin (10 µg), tetracycline (30 µg), streptomycin (10 µg), chloramphenicol (30 µg), trimethoprim-sulfamethoxazole (1.25/23.75 µg disk) ceftriaxone (30 µg), ciprofloxacin (5 µg), azithromycin (15 µg), and meropenem (10 µg). The inhibition zones were assessed and rated according to the CLSI as susceptible, intermediate, and resistant. The isolates were graded as MDR (Multiple Drug Resistant) and XDR (Extensively Drug Resistant) based on their resistance behavior towards three to four classes of antibiotics and five or six antibacterial groups, respectively (Parry and Threlfall 2008). Following CLSI guidelines (Wayne 2011), the isolates were also tested for ESBL phenotypes.

DNA extraction and ribotyping:

Genomic DNA was extracted by using WizPrep™ gDNA Mini Kit (Seongnam, South Korea), according to the manufacturer's instruction. High quality DNA (A260/A280 ratio was 1.8) was used as template for PCR amplification of 16S rRNA gene (Table 1) by previously reported primers (Edwards et al., 1989). Each 25µl PCR reaction mixture contained, 1µl DNA, 1µl reverse primer, 1µl forward primer, 9.5µl nuclease free water (ThermoFisher, MA, USA) and 12.5µl 2×PCR master mix (ThermoFisher, MA, USA). The PCR amplification was performed in thermocycler, Major Cycler system, Scientifix (Clayton, VIC, Australia). The PCR reaction was carried out as, initial denaturation at 95°C for 10 in followed by denaturation at 95°C for 30 s, annealing at 60°C for 1min and extension at 72°C for 1min for 35 cycles, final extension at 72°C for 1min. Amplified product was detected by electrophoresis using agarose gel containing ethidium bromide (Sigma-Aldrich, Missouri, USA) along with DNA molecular weight marker (Promega, Madison, Wisconsin, USA). Visualization of the gel was performed using benchtop ultraviolet illuminator.

PCR assays for resistance determinants:

Different antimicrobial resistance genes conferring bacterial resistance were amplified by using PCR amplification. PCR templates were prepared by boiling method as mentioned by Xiang *et al.* (2020). PCR was performed in 25µl PCR reaction mixture as described above. PCR reaction was carried out for 35 cycles including initial denaturation at 95°C for 5min, followed by denaturation at 95°C for

1min, annealing for 1min, extension at 72°C for 1min and final extension at same temperature for 10 min. Multiplex PCR was designed for the amplification of tetracycline resistance genes including *tet(A)*, *tet(B)* and *tet(C)* (Ng *et al.* 2001). Quinolone resistance genes *qnrS*, *gyrA*, *gyrB* and *parC* genes were amplified by previously reported primers (Eaves *et al.* 2002; Eaves *et al.* 2004; Lavilla *et al.* 2007). β-lactam antibiotics resistance genes like *ampC* (Pérez-Pérez and Hanson 2002), bla_{NDM-1} (Xiang *et al.* 2020), bla_{CTX-M} (Pagani *et al.* 2003) and aminoglycoside resistance gene *aac3* (Dubois *et al.* 2002) were amplifed by PCR reaction. List of all primers used in study are mentioned in Table 1.

The presence of class 1 integron integrase gene was tested by PCR, with the primers reported by Kheiri and Akhtari (2016). Variable regions in class 1 integron integrase gene were amplified by using primers complementary to 5' and 3' conserved region of class 1 integrase (Lévesque *et al.* 1995). The amplicon was purified and sent for sequencing to Macrogen Inc. Korea. Alignment of nucleotide sequences was performed with MEGA ver 7.2. The gene cassettes were selected and identified whose similarity with GenBank data was at least 95%.

Statistical analysis:

Statistical analysis was performed by using SPSS ver. 20.0 (Chicago, IL, USA), software. *P* value of ≤0.05 was considered statistically significant. The confidence intervals were calculated using the Clop-per-Pearson method (Cl. 95%).

Results:

A total number of two hundred (200) blood culture specimens of provisionally diagnosed enteric fever patients submitted at Department of Hematology, Chughtai's Laboratory, Lahore, Pakistan during 2019-20. Out of these, 180 (90%) blood culture specimens were yielded growth of *Salmonella* isolates confirmed by morphological, biochemical, serological, and molecular characterization. Among *Salmonella* cultures, 157 (87.2%) isolates were confirmed as *S.* Typhi and 23 (12.7%) isolates were confirmed as *S.* Paratyphi after molecular analysis. Majority of the cases were reported from Lahore city of Punjab, Pakistan in adult (15-45 years) age group (Table 2). Most of the patients had fever from last 4-7 days with varying clinical sign and symptoms including nausea, vomiting, diarrhea, fatigue, and constipation. Few patients also had previous record of antibiotics usage including azithromycin, ceftriaxone and meropenem (Table 3).

Antibiotic susceptibility testing:

In this study, 121 isolates (67.2%) were found to have multiple drug resistance (MDR) against different classes of antibiotics. Majority of MDR population consisted of *S.* Typhi while only six *S.* Paratyphi isolates were resistant to more than two or three antibiotics (Table 4). The frequency of extensively drug resistant (XDR) *Salmonella* isolates was 34.4% (62 isolates). All XDR isolates belonged to *S.* Typhi; no XDR case was observed among *S.* Paratyphi isolates. XDR isolates were resistant to first generation antibiotics, and in addition to quinolones and third generation cephalosporins. Resistance to ampicillin, trimethoprim-sulfamethoxazole, ciprofloxacin and chloramphenicol was recorded as 75%, 71%, 70% and 60%, respectively in *S.* Typhi and 43%, 30%, 65% and 26% in *S.* Paratyphi, respectively (Table 4). Only 1 isolate was resistant to azithromycin. However, all the isolates were susceptible to meropenem. Therefore, azithromycin and meropenem were the only option left for the typhoidal *Salmonella* cases in this study. One XDR isolate was resistant to azithromycin; however, it was susceptible to meropenem. Few isolates were intermediate to streptomycin, chloramphenicol, trimethoprim-sulfamethoxazole, and ceftriaxone antibiotics.

Ampicillin resistant gene (*ampC*) was observed in 86 (47.7%) isolates. *tet(A)* gene was detected in 39 (21.6%) isolates while *tet(B)* and *tet(C)* genes were not detected in any bacterial isolate. Chloramphenicol resistance gene *catA1* was found in 72 (40%) isolates. Contrary to the phenotypic expression of aminoglycoside antibiotic, genotypic expression of aminoglycosides was evaluated and found (18.3%) 33 isolates positive for *aac (3)-la* gene. Similarly, quinolone resistance genes, *gyrA* and *qnrS* was detected in 82 (45.5%) and 21 (11.6%) isolates, respectively. However, *gyrB* and *parC* genes were not detected in any isolate. Isolates exhibiting phenotypic expression of β-lactam resistance and ESBL production was in accordance with the amplification of bla_{NDM-1} and $bla_{CTX-M-15}$ in 2.2% (4 isolates) 0.5% (1 isolate) population, respectively (Figure1).

Integrase gene:

Analysis of class 1 and class 2 integron integrase genes were performed by PCR amplification and revelated relatively lower number of bacterial isolates were positive for integrase gene ($p > 0.05$). Almost 6.6% (12 isolates) bacterial population exhibiting class 1 integrase gene (*intI1*) while none of the bacterial isolate found positive for class 2 integrase gene (*intI2*). Gene cassette array of class 1 integron gene was amplified by PCR and their variable regions were sequenced. Among 12 class 1 integrase positive isolates, 6 isolates carried *sul1* and 4 isolates were positive for *dfrA7* gene cassette arrays.

Discussion:

Enteric fever is a systemic infection caused by *Salmonella enterica* with high rate of morbidity and mortality. Most of enteric fever cases have been reported in Asia; notifiable in Sindh province of Pakistan (Klemm *et al.* 2018). Typhoid fever has become a major health care concern in developing countries due to inadequate sanitary practices, unsafe potable water resource, overcrowding and lack of antimicrobial susceptibility testing data.

In current study majority of enteric fever infected patients were adult males with age more than 15 years. This phenomenon can be explained by considering the risk factors in male dominant society, where male mostly remain outside the home, eat outside and exposed to the *S.* Typhi infected patients (Qamar *et al.* 2018). Some Asian studies reported the incidence of enteric fever dominated among children aged <15 years (Qamar *et al.* 2018). Few studies reported the contamination of sewage lines with XDR *Salmonella* and large burden of *Salmonella* DNA in community water sources (Qamar *et al.* 2018). Reasserting the significance of the clean drinking water to lower the incidence of enteric fever cases in Pakistan. In current study, the incidence of *S.* Paratyphi was 8% while majority of patients were infected with *S.* Typhi (92%). This pattern is consistent in some other local studies as well (Shujat *et al.* 2016; Qamar *et al.* 2020).

This study reports the antimicrobial resistance and genotypic pattern of enteric fever patients from Pakistan. The percentage of MDR isolates among *S.* Typhi was quite high (97%) as compared to *S.* Paratyphi (7%). *S.* Typhi was found more prevalent, virulent and drug resistant as compared to *S.*

Paratyphi ($P < 0.05$). Previous studies suggest a proportional increase in MDR in *S.* Typhi from 20% to 50% in 1992 to 2015, whereas decline in MDR in *S.* Paratyphi cases was observed since 2004 to almost negligible levels (Klemm *et al.* 2018). Resistance to amoxicillin, trimethoprim-sulfamethoxazole and ceftriaxone in *S.* Typhi was 75%, 71% and 70%, respectively ($P < 0.05$). Resistance to ciprofloxacin, amoxicillin and ceftriaxone in *S.* Paratyphi was 65%, 43% and 39% respectively ($P < 0.05$).

Among *S*. Paratyphi isolates, highest antimicrobial resistance was observed against ciprofloxacin *i.e.,* 65%. Ciprofloxacin resistance pattern in *S*. Paratyphi observed in this study, was comparable with the previous reported research as well (Shujat *et al.* 2016; Mutai *et al.* 2018; Shah *et al.* 2020; Chen *et al.* 2020; Du *et al.* 2020). Ciprofloxacin remained a drug of choice in 1980s and 1990s epidemics of typhoidal MDR (Voysey *et al.* 2020). But unfortunately, extensive prescription, widespread use in animal husbandry and unregulated use of fluoroquinolones in developing countries, raised the fluoroquinolones non-susceptibility among *Salmonella* spp. up to 96.5% in 2015 globally (Holmes *et al.* 2016; Bhetwal *et al.* 2017). Consequently, fluoroquinolones lost its status as a drug of choice against typhoid infection and currently not recommended for typhoid treatment in Pakistan.

Progressive appearance of resistance against fluoroquinolones made third generation cephalosporins a drug of choice empirically in treatment of suspected enteric fever cases in Pakistan. Resistance against third generation cephalosporins were not reported until 2016, when a large outbreak of XDR typhoid infection was reported from Hyderabad, Pakistan (Aziz and Malik 2018; Klemm *et al.* 2018). Current study accounts 34% resistance against ceftriaxone in *S.* Typhi infected patients and 39% resistance in *S.* Paratyphi ($P < 0.05$). Previous local study reported up to 48% resistance of ceftriaxone against *S.* Typhi (Hussain *et al.* 2019).

Klemm *et al.* (2018) reported whole-genome sequence of *Salmonella* XDR isolates, representing a highly conserved H58 haplotype subclade. However, the IncY plasmid was extremely homologous to a homogenous group of plasmids broadly reported in other enteric bacteria around the world, suggested that the IncY plasmid may have been obtained through the conjugation from an *E. coli* which may present in individual's intestine. Additional genetic analysis found that the H58 subclade linked with XDR was utterly widespread in Pakistan and some other Middle East countries prior to the acquisition of the IncY plasmid present in Sindh (Klemm *et al.* 2018)**.** The emerging strain carried both the chromosome-integrated MDR enduring transposon possessing with extended beta-lactamase (ESBL) gene *blaCTX-M-15* mainly conferring the resistance to ceftriaxone and other cephalosporins of the third generation and the *qnrS* quinolone resistance gene which in combination with a chromosome *gyrA*-S83F mutation caused ciprofloxacin resistance (Ingle *et al.* 2019)**.**

The frequency of XDR S. typhi in our study was 34.4%. The XDR isolates were resistant to first line antibiotic, quinolones as well as third generation cephalosporins. In our study, only one isolate was resistant to azithromycin while all isolates were susceptible to meropenem. Thus, the treatment options left for typhoid fever are azithromycin and carbapenem (meropenem). Azithromycin is the only available oral antibiotic remaining for treatment of XDR *S.* Typhi. Resistance to azithromycin in *S.* Typhi has already been reported (Klemm *et al.* 2018) but its resistance in XDR *S.* Typhi isolates will be an added threat. Preliminary studies regarding potential treatment of SARS-CoV-2 with azithromycin (Gautret *et al.* 2020) resulted in irrational use of azithromycin in even suspected COVID-19 patients. Under current scenario, the unprecedented use of azithromycin in Pakistan (unpublished data) will enhance the chances of azithromycin non-susceptibility in XDR *S.* Typhi. If use of azithromycin will not be regulated, it will be a ticking bomb in which typhoid fever will eventually become untreatable in Pakistan.

Continuous rise in typhoidal XDR cases in Pakistan is not only a local health issue but also a global security concern. As there are many cases in which foreign travelers of Australia, Spain, Denmark, Italy, USA, UK and Canada were diagnosed with XDR *S.* Typhi infection from Pakistan (Chatham-Stephens *et al.* 2019; Howard-Jones *et al.* 2019; López-Segura *et al.* 2019; Wong *et al.* 2019; Procaccianti *et al.* 2020; Singanayagam *et al.* 2020). Hence there is a need of coordinated global efforts in controlling the spread of typhoid cases with the help of better diagnosis, effective surveillance, improving water and sanitary condition, and implementation of effective vaccine. Typhoidal conjugate vaccine (TCV) has a great potential to control the spread of typhoid cases. TCV vaccine globally introduced to vaccinate thousands of children in typhoid affected regions. Since 2019, Pakistan has become the first country to implement TCV vaccine in its national immunization program to control the further spread of typhoid.

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Conflict of interest:

The authors declared no conflict of interest.

Authors' Contribution:

IZ and AS designed and performed the experiments. AH and SC provided the bacterial cultures and patient's data. AA reviewed the manuscript and experimental results. All authors read and approved the final manuscript.

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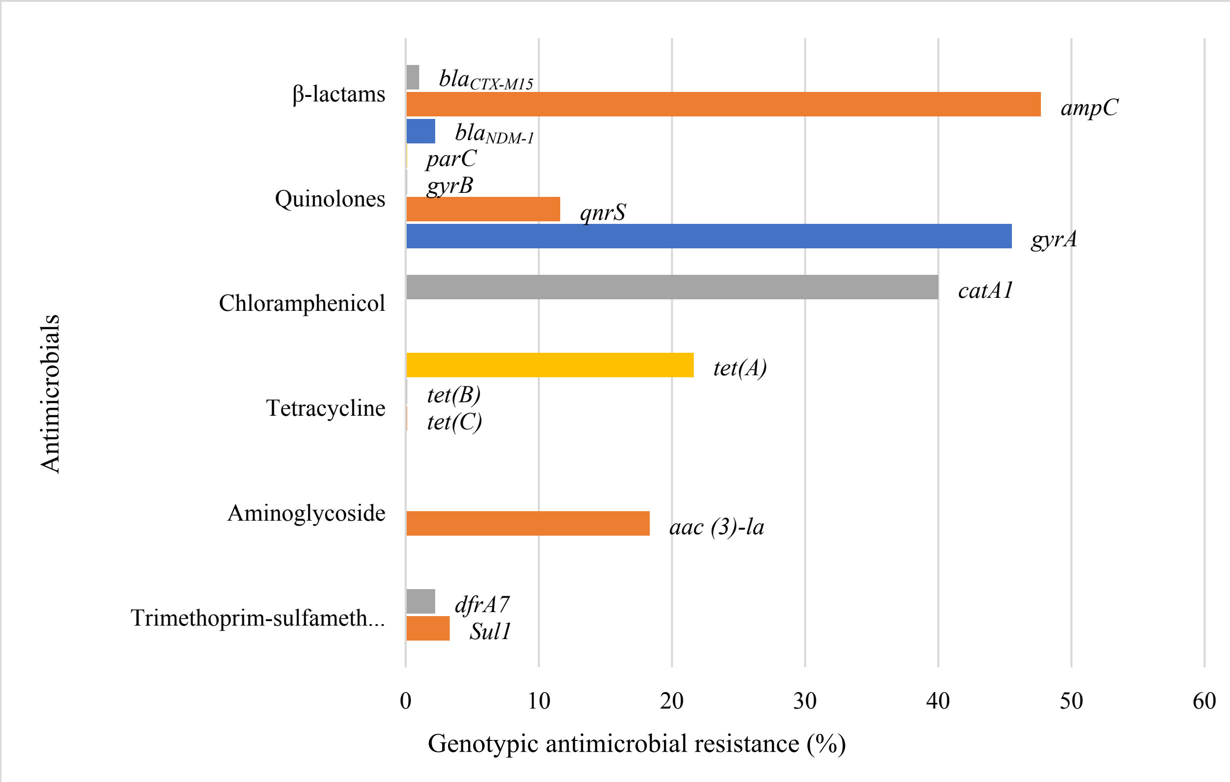

Figure 1: Genotyping of antimicrobial resistance in *Salmonella* spp. (n = 180).

Table 1: Primers used in this study

Target gene	Sequence (5'-3')	Reference
bla_{NDM-1}-F	GGCGGAAGGCTCATCACGA	(Xiang *et al.* 2020)
bla_{NDM-1}-R	CGCAACACAGCCTGACTTTC	
bla_{CTX-M}U1	ATGTGCACCAGTAARGT	(Pagani *et al.* 2003)
bla_{CTX-M}U2	TGGGTRAARTARGTSACCAGA	
ampC-F	AACACACTGATTGCGTCTGAC	(Pérez-Pérez and Hanson 2002)
ampC-E	CTGGGCCTCATCGTCAGTTA	
tet(A)-F	GCTACATCCTGCTTGCCTTC	(Ng *et al.* 2001)
tet(A)-R	CATAGATCGCCGTGAAGAGG	

tet(B)-F	TTGGTTAGGGGCAAGTTTTG	(Ng *et al.* 2001)
tet(B)-R	GTAATGGGCCAATAACACCG	
tet(C)-F	CTTGAGAGCCTTCAACCCAG	(Ng *et al.* 2001)
tet(C)-R	ATGGTCGTCATCTACCTGCC	
aac(3)-F	CATCATAGGAGGTTGTTTGATGTTATGGAGCAG	(Dubois *et al.* 2002)
aac(3)-R	TTATTATGGATCAATGTCGAAGTGCA	
qnrS-F	CCTACAATCATACATATCGGC	(Lavilla *et al.* 2007)
qnrS-R	GCTTCGAGAATCAGTTCTTGC	
gyrA (stgyrA1)	CGTTGGTGACGTAATCGGTA	(Eaves *et al.* 2002)
gyrA (stgyrA2)	CCGTACCGTCATAGTTATCC	
gyrB (stmgyrB1)	GCGCTGTCCGAACTGTACCT	(Eaves *et al.* 2004)
gyrB (stmgyrB2)	TGATCAGCGTCGCCACTTCC	
parC (stmparC1)	CTATGCGATG TCAGAGCTGG	(Eaves *et al.* 2004)
parC (stmparC2)	TAACAGCAGCTCGGCGTATT	
Integrase class I (*IntI. F*)	TCTCGGGTAACATCAAGG	(Kheiri and Akhtari 2016)
Integrase class I (*IntI. R*)	GTTCTTCTACGGCAAGGT	
Integrase class II (*IntII. F*)	CACGGATATGCGACA AAAAGGT	(Kheiri and Akhtari 2016)
Integrase class II (*IntII. R*)	GTAGCAAACGAGTGA CGAAATG	
Class I integron variable Region (F)	GGCATCCAAGCAAG	(Levesque *et al.* 1995)
Class I integron variable Region (R)	AAGCAGACTTGACCTGA	

Class II integron variable Region (F)	GATGCCATCGCAAGTACG AG	(White *et al.* 2001)
Class II integron variable Region (F)	CGGGATCCCGGACGGC ATGCACGATTTGTA	
pA (16S rRNA gene)	AGAGTTTGATCCTGGCTCAG	(Edwards *et al.* 1989)
pH (16S rRNA gene)	AAGGAGGTGATCCAGCCGCA	(Edwards *et al.* 1989)

Table 2: Characteristics of culture positive patients and isolates

Gender	
Male	125 (69.5%)
Female	55 (30.5%)
Age group (Years)	
<15	31 (17.2%)
15-45	92 (51.1%)
>45	57 (31.6%)
Species	
S. Typhi	157
S. Paratyphi	23

Table 3: Clinical characteristics of typhoid fever patients

Fever (Reporting days)	
4-7	115 (63.8%)
8-14	59 (32.7%)

>14	6 (3.3%)
Clinical sign & symptoms	
Fatigue	120 (66.6%)
Nausea	32 (17.7%)
Diarrhea	26 (14.4%)
Vomiting	18 (10%)
Antibiotic usage history	
Ciprofloxacin	18%
Ceftriaxone	12%
Ampicillin	9%
Azithromycin	3%
Meropenem	1%

Table 4: Antimicrobial susceptibility pattern of *S.* Typhi and *S.* Paratyphi (*n* = 180).

Antibiotics	*Salmonella* Typhi *(n = 157)*			*Salmonella* Paratyphi *(n = 23)*		
	R	**I**	**S**	**R**	**I**	**S**
AMP	119 (75%)	0 (0%)	38 (25%)	10 (43%)	0 (0%)	13 (56%)
TET	85 (54%)	0 (0%)	72 (46%)	7 (30%)	0 (0%)	16 (70%)
STR	62 (39%)	2 (1%)	95 (60%)	8 (34%)	0 (0%)	15 (65%)
CAP	95 (60%)	4 (2%)	58 (37%)	6 (26%)	0 (0%)	17 (74%)
SXT	112 (71%)	5 (3%)	40 (25%)	7 (30%)	0 (0%)	16 (70%)
CRO	54 (34%)	8 (5%)	95 (60%)	9 (39%)	0 (0%)	14 (61%)
CIP	110 (70%)	0 (0%)	47 (30%)	15 (65%)	0 (0%)	8 (35%)
AZT	1 (0.6%)	0 (0%)	156 (99%)	0 (0%)	0 (0%)	0 (0%)
MEM	0 (0%)	0 (0%)	157 (100%)	0 (0%)	0 (0%)	0 (0%)

AMP (ampicillin 10 μg), TET (Tetracycline 30 μg) STR (streptomycin 10 μg), CAP (chloramphenicol 30 μg), SXT (trimethoprim-sulfamethoxazole 1.25/23.75 μg), CRO (ceftriaxone 30 μg), CIP (ciprofloxacin 5 μg), AZT (azithromycin 15 μg), MEM (meropenem 10 μg), S (susceptible), I (intermediate), R (resistant).